PASHURAKSHAK
IDEA SUBMISSION

TEAM PASHURAKSHAK

PashuRakshak

Livestock Disease Early-Warning System

An offline-first Android + web ecosystem that helps farmers report sick animals, vets respond faster, and governments detect outbreaks before they spread.

AI Risk Scoring

Outbreak Clustering

Offline-First

QR Passports

Cloud Sync

“Protecting livestock, protecting livelihoods”

A complete digital ecosystem for livestock health monitoring, outbreak detection and veterinary management across rural India.

The Why

The Problem & Our Solution

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Rural livestock health is reactive today. PashuRakshak makes it proactive.

Today's Reality

- Farmers have no quick, reliable way to report a sick animal.
- Vets face unsorted caseloads with no triage or priority.
- Outbreaks spread silently; response is always reactive.
- Patchy connectivity and paper records slow every step.

The PashuRakshak Way

- Report sick animals with a photo and GPS in seconds.
- AI risk scoring + a priority queue routing vets to urgent cases.
- Automatic outbreak clustering with instant, targeted alerts.
- Offline-first design that syncs the moment the network returns.

One connected platform — from the farmer's field to the District Magistrate's dashboard.

Why PashuRakshak

The Gap Today & Our Edge

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Is anything like this already out there? Here is what exists — and what is missing.

Existing solutions — and their limits

- Paper registers & word-of-mouth — no data, no alerts.
- Govt reporting portals — slow, one-way, not field-ready.
- Generic telemedicine apps — built for humans, not livestock.
- Spreadsheets / WhatsApp groups — no triage or verification.
- Single-purpose tools — vaccination OR insurance, not connected.

How PashuRakshak compares

Capability	Today	PashuRakshak
Offline-first field reporting	Rare	Yes
AI triage + vet verification	No	Yes
Automatic outbreak clustering	No	Yes
Farmer, vet & govt in one loop	No	Yes
Verifiable QR + audit trail	No	Yes
Free, multi-role ecosystem	Partial	Yes

Our edge: one connected, offline-first ecosystem — AI-assisted but always vet-verified, and built for rural India.

How It Works

End-to-End Workflow

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A single report flows through scoring, outbreak detection, a verified vet visit and government action.

if risky

1 Report

Photo, GPS & symptoms captured

2 AI Risk Score

Scored 0-100
Low / Medium / High

3 Cluster Check

7-day · 5 km ·
3+ reports

4 Vet Response

Priority queue +
QR-verified visit

5 Govt Action

Risky case:
alert + ack.

Offline-first

Reports queue locally and sync in the background.

Verifiable & audited

QR codes are HMAC-signed; every action is audit-logged.

Explainable AI

Each score shows the symptoms that drove it.

POWERED BY Groq LLM · Firebase FCM · Backblaze B2 · Postal API **BUILT WITH** Kotlin · Jetpack Compose · Node.js · Express · Turso · React 19

Who It Serves

One Platform, Four Experiences

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Purpose-built interfaces for every stakeholder in the livestock ecosystem.

Farmer App

Android

- Report sick animals with photo, GPS & symptoms.
- Instant AI risk level — online or offline.
- QR animal passports + vaccination tracking.

Vet App

Android

- Priority queue — red alerts first.
- Field tools: examine, notes, sample requests.
- Explainable AI + “3+ cases within 5 km” warnings.

Government Portal

React Web

- Live dashboard of reports, cases & clusters.
- Vet approval with pincode, then district routing.
- Outbreak maps + official alert acknowledgment.

Vet Registration

React Web

- Email OTP verification + secure PIN setup.
- Professional application with coverage areas.
- Real-time application status tracking.

The Intelligence

AI Risk Scoring & Outbreak Detection

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Turning raw field reports into ranked, actionable warnings.

AI Symptom Risk Scoring

Every report is scored 0–100 by an LLM using weighted symptoms (fever, lethargy, appetite …).

- **HIGH 60+** • immediate vet alert
- **MEDIUM 30-59** • routine triage
- **LOW < 30** • self-care guidance

Model: openai/gpt-oss-120b via Groq API.

Automatic Outbreak Clustering

Reports are grouped geographically to catch outbreaks early — no manual effort required.

7-day	5 km	3+
window	radius	reports

Triggers push alerts to nearby vets & the government portal.

Trust Layer

Verifiable records that make the data trustworthy.

- **Signed QR records**
- **Vet-verified reports**
- **Immutable audit log**

The Vision

What's Next — Feature Roadmap

PLANNED

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The capabilities we are designing on top of today's platform.

Advanced Intelligence	Photo lesion detection · learning from vet outcomes · outbreak forecasting · cited (RAG) advisories · regional-language voice · CI safety-eval (never prescribes)
Farmer Reach	Vet teleconsultation · treatment & withdrawal tracking · herd records & productivity · reminders via push/SMS/WhatsApp · localization (Bhashini) · accessibility · community alerts · insurance export
Vet Workflow	Smart assignment (distance, caseload, specialty) · SLA escalation · travel-time routing · structured clinical forms · e-prescriptions · cold-chain inventory · lab integration · analytics
Government & Oversight	H3 geospatial heatmap · quarantine zones · movement e-permits · vaccination campaigns · official PDF reports · multi-tier roles · escalation & audit trail
Integrations & Hardware	National livestock ID linkage · WhatsApp reporting bot · maps & geocoding · IoT sensors

Every capability keeps the same guarantees: AI defers to a vet, and every action stays auditable.

Impact · Why Now · The Ask

Early detection saves livestock, livelihoods and money

Expected Impact — Year 1

10,000+	farmers onboarded across 3 districts
500+	symptom reports filed per month
< 24 hrs	response time to high-risk alerts
15+	outbreaks detected early per month

Why Now

- Aligns with the National Livestock Mission & Digital India.
- Rural smartphone + WhatsApp penetration is at an all-time high.
- Livestock is a lakh-crore economy with huge, preventable losses.

DELIVERY ROADMAP

P1	P2	P3	P4	P5
Harden & localize	Smarter triage	Farmer reach	Predictive & spatial	Ecosystem & IoT

Let's bring PashuRakshak to every farmer

One connected, offline-first, AI-assisted — yet always vet-verified

Team PashuRakshak